

G54B1 - technical appendix to the first external-authoring case

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Recorded result: G54B1_COLD_AUTHORIZING_INCOMPLETE_NATIVE_OFFER_BINDING
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Status: Historical evidence and analysis. **Final acceptance:** INCOMPLETE.

This appendix accompanies [the case presentation](#), [the neutral machine reader](#), [the claim map](#), [the trial ledger](#), and [the exact source map](#). The .pttx is the editable presentation. All named candidate source files in evidence/recorded/ are **inert historical data**. Their publication neither installs a domain nor grants admission.

Abstract

G54B1 asks whether a fresh external model developer can construct a new Radiolaria domain from a bounded authoring kit without controlling the kernel, examiner, or admission. The recorded trial contains an initial submission and three corrections through a restricted broker. Its final candidate performed a two-party football booking scenario with External DRS and native Work: after an explicitly permitted schedule revision, three synthetic sessions were recorded by one mock effect, with a matching receipt and readback.

Final acceptance remained **INCOMPLETE**. The preserved evidence exposed an interface inconsistency: the producer’s semantic result, its later provenance-bearing publication, and the consumer’s status-enriched context were required to be the same complete value. Independent inspection verified exact relations between these different representations and their source events. The case shows useful externally authored candidate functionality and an informative contract failure. It does not close Gate 5 or demonstrate autonomous self-installation.

Reading key and evidence levels

In source citations, S0:P means member g54b1_return/P of the recorded archive. The local publication copy is evidence/recorded/P, except an original .py suffix becomes .py.txt; bytes are unchanged. [SOURCE_MAP.json](#) gives every original path, published path, byte count, SHA-256, and suffix transformation. S1 is the later [independent contract review](#), separately identified and not a native pass. Source identifiers used below:

ID	Evidence	What it establishes and its limit
S0	Original 7,928,207-byte archive, SHA-256 9c383d342b63c3ca54cddc51c37eae0baac53ccc72bf49fd9a1bb8ce67f2d422	Recorded program, inputs, examiner, feedback, native runs, and observer. A recording is not a new replay.
S1	Independent review, SHA-256 87152f7f44df0aa3995fd4edd960e44f55c086368abbb133d23607fa6b543e18	Independently checked scalar combinations, E/B/C, six historical signatures, registry and receipt. Candidate code and model were not rerun.
S4	Publication source check supplied with the directive	Archive/manifest identity and selected file hashes; no candidate execution or new signature review.

OBSERVED below means recorded execution or direct archival facts; REVIEW_FINDING identifies a separate comparison; INTERPRETATION is an argument with stated scope; FUTURE is an unperformed design or trial. The label is a kind of claim, not a numerical confidence score. See the machine-readable claim map for the source binding of T01-T12.

A. Scope and trial boundary

A.1. Three results that must remain distinct

Functional observation. The model generated executable new domain code. In the last saved confined run, native Work on the venue side computed an offer from a supplied inventory. A different Root-side Work consumed the obtained offer and status context. A separately bounded synthetic mock booking changed the venue's registry; the result was read back, and repeat and verify returned the same receipt. These are recorded, source-bound operations, stronger than a text-only demonstration.

Acceptance outcome. The frozen independent native examiner stopped at venue_offer_consumption (EXAMINER_PRIVATE/examiner.py:164). The full P01-P10/A-H matrix, runtime live Gemini series, and the two final supplied proofs did **not** run after that failure. The recorded outcome remains G54B1_COLD_AUTHORIZING_INCOMPLETE_NATIVE_OFFER_BINDING; no Gate 5 completion, owner source admission, G55, or G56 is implied. Source: S0:FINAL_RESULT.json /result, /independent_native_exam, /live_gemini_series, /final_two_supplied_proofs.

Contract finding. The author's last correction identified a causal inconsistency in the shared-offer representation. The later independent reviewer checked actual producer Work, published bytes, status, and consumer Work in both EXACT and SHIFT. The examiner compared the entire later body to the earlier semantic result and the same body to the later status-enriched consumer input. The finding is about this declared one-producing-Work interface, not a proof that no alternative architecture can exist. Sources: S0:AUTHOR_VISIBLE/HOW_TO_BUILD.md:160-169,209-210; S0:EXAMINER_PRIVATE/examiner.py:155-169; S0:AUTHOR_TRIAL/attempt_03/candidate/candidate_domain/CORRECTION_03.md:78-125; S1:\$5.

The publication's architectural formula is: *A system can expand what it does without giving the author of the expansion control over the rules that govern it.* This bounded trial illustrates a separation of creation, execution, review, and admission. It does not certify protection against an administrator, an undisclosed service path, or every possible hostile candidate. The author was fresh to this thread and was not passed prior target code; pretrained knowledge cannot be ruled out. The generated candidate remained installable_now=false and quarantined, as its own final note says.

A.2. Chronology and the meaning of “first”

Record	Nature	Outcome	Relation to this publication
G54R	Earlier Gemini gemini-2.5-flash cold-author trial; initial plus three corrections	TRANSPORT_RECOVERED_COLD_A UTHOR_TRIAL_INCOMPLETE; final peers failed before native Work/Root	Preserved predecessor, not one of G54B1’s four submissions.
G54B	Astra access-boundary probe; no football task delivered	AUTHOR_BOUNDARY_UNVERIFIED ; author trial not started	A predecessor diagnostic, not an accepted authoring trial. Earlier standalone command/exec discrepancy was preserved, not declared repaired.
G54B1	New recorded environments=[] Astra authoring thread	Initial submission + three corrections; functional run; final native acceptance INCOMPLETE	The first standalone showcase selected by the owner , not the first model-authoring attempt in the project.

The predecessor states are copied verbatim from `S0:SUPERVISOR_EVIDENCE/G54R_FINAL_RESULT.json /RESULT, /TRIAL, /FOOTBALL_NATIVE_RESULT` and `S0:SUPERVISOR_EVIDENCE/G54B_FINAL_RESULT.json /RESULT, /AUTHOR_TASK_DELIVERED, /FIRST_BLOCKER`. `S0:SUPERVISOR_EVIDENCE/predecessor_preservation.json` records 147 unchanged earlier input bodies. [The trial ledger](#) maintains separate identities and missing/unknown fields. Four G54B1 submissions are one bounded author cycle, not four independent authors or a statistical sample. This publication is dated after the trial; editorial QA is a third axis and cannot retroactively change the trial result.

B. Author-visible inputs and access surface

B.1. The actual assignment

The model was configured as `gpt-6-astra, effort ultra, service tier default, with environments=[]`; the separate fresh author thread received `AUTHOR_SYSTEM.txt, HOW_TO_BUILD.md, WHAT_TO_BUILD.md, CANDIDATE_OUTPUT.md`, an approved source snapshot/interface index and the bounded tools. The exact frozen input hash is `972ca734b2e19aedec39e79ffc5cb70f37ebf9425d9416bcd33026d74e084ecd` (`S0:AUTHOR_TRIAL/freeze.json /input_sha256`). The author-facing generic `HOW` hash is `6636167b75d8fec5786e80cb5ce3027c02748d58d5d63765655b1e451d60dea3`; task-specific `WHAT` hash is `dbcdfeeaca9405687dd30bba33a72a510729d606c7d99ee52a5f390d426963a1`. The frozen examiner hash is `874b4b7a82c15bc2212404cceef182519ef16681bb9a85b0addf6f6f336ce2372`. The recorded owner `HEAD 76a6952c9d0f31c976d2501c5641457a20de153a` is historical context, not a required current checkout.

`HOW` gives `Root/Work/Host/Firewall/DRS` rules, a specific available interface profile, and a numerical donor that computes $3 \times x + 2$. It does not supply a completed football solver or the final booking. `WHAT` declares the synthetic football scope, inputs, hard conditions, permitted operations, exactly scoped January 8 relaxation, and receipt/readback requirements; it forbids guessing missing constraints and real bookings. Separating these documents makes a new domain’s meaning task-specific while leaving the common extension process visible. `CANDIDATE_OUTPUT.md` mixes common return-package structure with football-specific fields; the later review recommends explicitly labelling those fields a

task profile. That editorial flaw is not itself evidence of a planted target answer. Sources: S0:AUTHOR_VISIBLE/HOW_TO_BUILD.md:1-44,160-169; S0:TASK_VISIBLE/WHAT_TO_BUILD.md:1-36,65-123; S0:AUTHOR_VISIBLE/CANDIDATE_OUTPUT.md; S1:\$3.

The effective configuration recorded a public-root read surface, read-only source access, network disabled, no shell/unified exec, no web search, no memory use/generation, no subagents, and no image/browser tools. A restricted broker exposed observable source read/search, candidate read/write/patch/inspect, and submit operations. The four author turns produced 429 **broker operations**: 184 candidate reads, 82 approved-source reads, 56 candidate patches, 46 approved-source searches, 44 candidate writes, 13 candidate inspections, and 4 submissions. The separate diagnostic used four dynamic tool calls. Neither count is provider inference requests; the app server did not expose that count. Sources: S0:AUTHOR_TRIAL/session/effective_config.json; S0:AUTHOR_TRIAL/session/receipt.json /tools, /turns_sent; S0:FINAL_RESULT.json /author_dynamic_tool_calls, /provider_inference_request_count; S1:\$2.

The initial diagnostic of environments=[] and broker operations has narrow value: its recorded calls and denials are evidence about that observed surface. The old standalone RPC permission discrepancy remained unexplained and was not repaired by this trial. Opaque platform instructions were declared trusted but not disclosed or hashed (S0:AUTHOR_TRIAL/freeze.json /opaque_platform_instructions). These are material limits on any absolute isolation claim.

B.2. Who held which power

Area	Recorded responsibility	What the model author could do
External author	Create and correct candidate files through the broker; receive selected feedback	Propose complete inert packs, inspect allowed sources, report a blocker.
Broker and supervising reviewer	Enforce observed tool surface, freeze submissions, supply feedback and choose runner invocation	Not supplied as editable candidate code. Access claims are limited to saved configuration/journal.
Confined runner	Execute frozen candidate under reviewed conditions and save native and observer records	Author did not run arbitrary host commands or directly install into owner.
Native Radiolaria	Local VenueRoot and ActivePlanetRoot decisions, Work, transport, Host/effect and provenance	Candidate used exposed seams; a foreign pointer/signature was not a local grant.
Examiner and supervisor	Independent acceptance predicate, observer, corrections and stop	Frozen examiner could fail the candidate; author’s note could not rewrite it.

This decomposition matters causally. Broker restrictions are not kernel invariants. The runner’s observations are not the author’s authority. Native Root acceptance of a bounded source or mock action is not a domain-level admission certificate. The wrong examiner comparison is a kit/examiner mismatch, not proof that native Root silently changed its rules. Sources: S0:AUTHOR_TRIAL/initial_input.json, AUTHOR_TRIAL/freeze.json, AUTHOR_TRIAL/attempt_03/REVIEW_DECISION.json; S0:FINAL_RESULT.json /native_kernel_guard_examiner_changes, /supervisor_target_code_edits.

C. Candidate generations and review feedback

C.1. Four frozen versions in one thread

Submission	Candidate manifest SHA-256	Actual observed result and first blocker	Feedback and correction boundary
attempt_00 / initial	68e3c14dab5824086cfce831f8bb59255b471a7d07af03bdb2b6341e85809e32	Requester exited before its Work; public DRS semantic-address operation refused with <code>drs_secret_payload_forbidden</code> . Venue Work and source/publication reviews had completed. This error code alone establishes neither a discovered secret nor a leak.	CORRECT; author adjusted representation of public DRS data in candidate code. Shift/booking were not reached.
attempt_01 / correction 1	712a382be898900eb2e17834abb1bb08c384b9c7d5922d3b1715c016527e4e7a	Both peers exited rc0, but requester returned REFUSED/json_key before FETCH/Work. Two legitimate composite highwater keys were 173 characters under a public JSON key limit of 128. No booking/effect.	CORRECT; saved-state diagnostic plus static localization. A normal process exit was not domain acceptance; public limit was not relaxed.
attempt_02 / correction 2	74a83d8f6d75e03009da809dd7d736737a68bf4e784e07a6673ea5afece7caa2	Exact and approved-shift scalar phases worked; book performed the mock effect, then mappingproxy serialization failed while writing <code>native_dispatch.json</code> . The registry already had three reservations and one idempotency entry; overall run rc1.	CORRECT; supervisor explicitly preserved the post-effect state and asked for candidate-only evidence/cleanup repair. Separate offer-binding and Root-claim diagnostics remained.
attempt_03 / correction 3	13986b2ff3f3dc7b1b77a9f884ff4521ee6f4cb0b10d44203403f42d8913d1aa	Both peers rc0; exact/shift, bounded mock booking/repeat/verify, receipt/readback. Native acceptance stopped on examiner line 164; shared-offer supplied diagnostics still refused expected inputs.	STOP. Correction quota exhausted; neither a fourth correction nor an alternate checker was applied. The author's correction note retained BLOCKED_SHARED_OFFER_IDENTITY_CONTRACT.

Version pins come from `S0:AUTHOR_TRIAL/attempt_00..03/FROZEN.json` /manifest_sha256; outcomes from each `REVIEW_DECISION.json`, `native_review.md`, and supervisor evidence. Each feedback text was delivered to the next author turn as a bounded `feedback_input.json`. The final candidate manifest covers 26 files, including 15 Python files and 2,419 physical Python lines including comments, tests, and blank lines. Lines of code are descriptive inventory, not a correctness metric. Final count is the recomputation in S4. Candidate snapshots, reviews and corrections are preserved in `evidence/recorded/AUTHOR_TRIAL/attempt_00..03/`.

C.2. Authored code versus executed checks

The candidate contained its own tests/test_*.py sources and README assertions. The unchanged peer runner did not expose unittest discovery for those tests; they were source-reviewed, **not executed** as proof. The supervising runner executed confined two-party stories, independent scalar checks, narrow public supplied diagnostics, and ultimately a frozen examiner prefix. attempt_03 did pass an exact three-hash Root-claim diagnostic, but that is not a substitute for the refused shared-offer predicate or the two final supplied proofs. The complete adversarial matrix did not run. Sources: S0:FINAL_RESULT.json /authored_unit_tests, /root_claim_diagnostic, /independent_native_exam; S0:AUTHOR_TRIAL/attempt_03/REVIEW_DECISION.json; S0:AUTHOR_TRIAL/attempt_03/candidate/candidate_domain/CORRECTION_03.md:127-137.

The author’s report of an unresolved contradiction was a candidate observation, not a self-issued override. The later reviewer tested it against original inputs, published bytes, saved STATUS and actual Work. Earlier errors remain candidate defects rather than being retrospectively blamed on the examiner.

D. Last recorded football run

D.1. Controlled request and an independently checked choice

The narrative “a team asks for three sessions” is a **paraphrase of controlled synthetic events**, not a verbatim live user quote or real venue appointment. The test task sets three 90-minute sessions on 7, 8 and 9 January 2027 in Europe/Tirane, natural grass and full-sized pitch, and a budget of 7,000 minor EUR units (€70.00). The first request fixes all three times; the inventory cannot satisfy a complete lawful combination. The actual EXACT result on both sides is NEEDS_CONFIRMATION, with no booking. A later request revision carries explicit owner_approval_ref=consent:jan8-only and permits a +30-minute move **only on 8 January**. The inventory includes a cheap artificial field (price 1 minor unit) that cannot meet the hard natural-grass requirement, and field:elm, natural/full-size at 2,100 minor units per slot. A 30-minute occupied interval blocks the original January 8 time on field:elm. Sources: S0:TASK_VISIBLE/WHAT_TO_BUILD.md:12-63; S0:SUPERVISOR_EVIDENCE/p7_candidate_1244df5d/worker/publisher/evidence/{exact,shift}/request.json, .../shift/inventory_input.json, and both requester report.json files.

Session	Exact request, local time	Selected after consent, local time	Field and price
January 7	16:30-18:00	16:30-18:00	field:elm, natural/full-size, 2,100 minor EUR
January 8	11:00-12:30	11:30-13:00	field:elm, natural/full-size, 2,100 minor EUR
January 9	11:00-12:30	11:00-12:30	field:elm, natural/full-size, 2,100 minor EUR

Total: **6,300 minor EUR = €63.00**, within **7,000 minor EUR = €70.00**. Independent review enumerated the actual saved intervals: zero complete candidates for the unapproved exact request; one when the permitted January 8 revision is applied; one in the actual revised request. The winning tuple matches native offer and requester report. This finite verification of the saved example does not replace the unrun variation and adversarial matrix. Source: S1:\$4;

S0:SUPERVISOR_EVIDENCE/p7_candidate_1244df5d/worker/publisher/evidence/shift/venue_work.json /inputs, /outputs; .../requester/evidence/shift/report.json /slots, /total_minor.

D.2. Native work, transport and separate decisions

VenueRoot generated a domain result by native Work from its own controlled request and inventory. A source review followed it. The candidate then formed a provenance-bearing body and manifest/pointer. The requester observed signed STATUS, resolved a bounded External DRS descriptor and received matching body bytes, passed local checks, and invoked separate ActivePlanetRoot Work on the actual received content plus its status projection. The final requester Work output matches its saved result report after removing the report-only native_refs. S1:\$5 independently checked Work IDs, source review ID, byte identity of body on both peers, invocation inputs and outputs, and the projection from saved STATUS. The review also checked six historical POINTER/STATUS/RELEASE Ed25519 signatures for EXACT and SHIFT against saved public bootstrap keys. These checks bind archived bytes to test keys; they do not certify real-world signers or independently replay every Root event. Sources: S0:SUPERVISOR_EVIDENCE/p7_candidate_1244df5d/worker/publisher/evidence/shift/{venue_work,source_review,offer_body,manifest,descriptor,signed_status}.json ; requester .../shift/{exchange_fetch,status,requester_work,report}.json; S1:\$5-6.

In the recorded system, an external DRS pointer transports evidence; it does not grant authority across Roots. The venue's Root decides its publication and synthetic booking; the requester's Root decides its local import/use. The common source identifiers and cryptographic records provide provenance, not a merged super-Root or a shared grant. A live runtime language-model actor interpreting open-ended customer requests was **not** run in G54B1 (gemini_runtime_calls=0).

E. Mock effect, idempotency and the earlier post-effect crash

E.1. The final run's bounded state change

For the final CONFIRM MOCK BOOKING, the saved venue registry has exactly three reservations and one idempotency entry. The registry bytes equal the saved booking readback bytes: both have SHA-256 1e0d2d310d7032ae3a8471e771f54cf6dabc1616682237527a763c94361d7300. The native receipt artifact is effect_receipt_v01:16d2611a5c85f250f59ad56e38cca6ce24922af4f56efe3b0791169ce30c2966. Venue and requester BOOK, REPEAT and VERIFY reports point to the same receipt. The saved trusted-parent observer logged one total publisher dispatch and one return from execute_bound_effect_v01 in that final run, with no requester dispatch/effect return. It did **not** split its totals into measured per-event intervals; the candidate's zero-increment counters on repeat/verify are distinct observations.

Sources: S0:SUPERVISOR_EVIDENCE/p7_candidate_1244df5d/trusted_parent_receipt.json, .../worker/publisher/state/registry.json, .../publisher/evidence/book/{receipt,readback,report}.json, .../publisher/evidence/{repeat,verify}/report.json, and requester counterparts; S1:\$4.

This was a **synthetic, local MOCK** effect in the bounded test. It was not a real pitch booking, contact, payment, cloud purchase or QPU call. Receipt reuse in this recorded case does not certify exactly-once behavior under every OS crash, network fault or external provider. Rechecking history is not a fresh permission to dispatch an old consumed action.

E.2. Attempt 02: the effect preceded its missing evidence file

The preceding candidate illustrates why effect and evidence must be traced separately:

1. The exact and explicitly revised offer phases reached legitimate scalar answers and actual Venue/Requester PURE Work.
2. In the BOOK event, the trusted parent recorded returns from native_dispatch_current_action_v01 and execute_bound_effect_v01. The saved registry already held three reservations and one idempotency entry.
3. Candidate effect.confirm then attempted to serialize a live mappingproxy through the public plain converter while saving native_dispatch.json; TypeError('mappingproxy') interrupted the evidence write. The candidate's later mutation counter update and cleanup had not run; a displayed mutations=0 at that point was not evidence of zero effect. Final receipt.json and native_dispatch.json did not complete.
4. Requester SOURCE_UNAVAILABLE_PEER_CLOSED and BrokenPipeError were downstream after the publisher failure. The supervisor correlated the registry and trusted parent feedback rather than declaring no effect or replaying it for the illustration.

Sources: S0:AUTHOR_TRIAL/attempt_02/REVIEW_DECISION.json /feedback; .../attempt_02/native_review.md; S0:SUPERVISOR_EVIDENCE/p7_candidate_0d000638/ (see its command receipt, parent observer and registry via SOURCE_MAP); S0:AUTHOR_TRIAL/attempt_03/candidate/candidate_domain/CORRECTION_03.md:11-26,28-63. The third correction proposed one-way ABI projection, saved terminal history before a larger audit and explicit uncertain outcome/readback. The final confined run succeeded for its saved inputs. This is **not** evidence that the runtime independently healed every crash; the supervising reviewer found and returned the earlier mismatch.

F. Exact E → B → C correspondence and frozen inconsistency

F.1. Define values before comparing them

All three names denote **decoded JSON values**. offer_json itself is a JSON string field; a comparison must decode it or compare canonical bytes of values at the same layer:

Name	Derived from actual saved source	Contents and causal time	SHIFT canonical SHA-256
E	decode(Venue Work outputs.offer_json)	Semantic offer computed from request and inventory by the producing Work.	1b99b5bca12497294cad821abbe f4521448de12d992e09edd2d504 08c0cac033
B	Decoded publisher/evidence/shift/offer_body.json, also received body	offer=E plus source Work/review identities, request, time envelope and provenance; formed after Work and review.	b92ee183c725b2052e29cd1840d 195fb3c5bc6b0d78984a1d1339f 5285e61ab3
C	decode(Requester Work inputs.offer_json)	{body:B, source_projection:...}; projection derives from received body and an authenticated later STATUS observation.	feb77a169eeb0470a78aa8bb6c5 c5d69e0afe13a6813e41042f2bf 1ac2fa55a1

The exact relations checked in both EXACT and SHIFT were B.offer == E, C.body == B, published-body bytes equal received-body bytes, B.source_work_ref == actual Venue Work artifact_id, B.source_review_ref == saved source-review decision_id, and C.source_projection == independently reconstructed projection(B, saved STATUS). The saved SHIFT Work ID is

work_results:2f720706a5e13c19d57c1755a1280eaac30d0ec12ade947dcf1982096aefe40f; review ID is cea16f779c17d64c29ed4ccebcb800b5bfca7d6954d0bfd66b7fdec440e9d0e39. Actual Work input/output proof relations and consumer result/report relation were also checked. Crucially, the expected consumer value came from original producer, received body and STATUS relationships in S1, **not** from assuming the candidate's own requester.inputs were the expected oracle. Different complete-value hashes alone would establish no semantic binding; field-by-field origins and actual event IDs do that. Sources: S0:.../publisher/evidence/shift/venue_work.json /outputs/offer_json, /artifact/artifact_id; .../source_review.json /result/decision_id; .../offer_body.json; requester .../shift/{offer_body,status,requester_work,report}.json; S1:\$5.

F.2. What the frozen requirements simultaneously demanded

HOW_TO_BUILD.md:160-169 requires the sequence **actual source inputs → native Work → current source review → body** containing the resulting Work/review IDs. It also requires those body fields at lines 209-210. The original examiner's check_native at lines 161-169, however, calls native_result on the Venue Work and at line 164 requires its **whole** offer_json **output** to equal canonical serialization of the later **whole** offer_body.json. At line 165 it gives canonical whole offer_body as the expected requester offer_json input. The public supplied validator checks actual invocation input equality; the saved requester consumed C, not B. Both comparisons refused at that interface, while native_result against each side's own recorded inputs and the separate exact three-hash claim diagnostic succeeded. The latter successes do not satisfy the examiner's mismatched asserted relationship. Sources: S0:AUTHOR_VISIBLE/HOW_TO_BUILD.md:160-169,209-210; S0:EXAMINER_PRIVATE/examiner.py:155-169; S0:AUTHOR_TRIAL/attempt_03/native_binding_diagnostic.json /checks; S0:FINAL_RESULT.json /first_unresolved_predicate.

For this declared single producing Work, demanding a body containing IDs of the Work and its subsequent review to be the earlier Work output creates a downstream/self-reference. The requester's later verified status projection also cannot be silently treated as bytes already present in the earlier published body. An explicit versioned DomainResult → PublishedEvidence → ConsumptionContext contract can require exact payload preservation and causal proof for each added layer. This is a **review finding and design recommendation**, not permission to silently loosen checks, patch the archived examiner, or mark the old result PASS. The author recorded this blocker in CORRECTION_03.md; independent S1 review reproduced exact relations rather than trusting that note alone.

F.3. Signatures and their scope

S1 independently recalculated body/manifest and six Ed25519 signature relationships: POINTER, STATUS and RELEASE for each of EXACT and SHIFT. It used the saved bootstrap public key, verified purpose/domain separation, manifest and value hashes, owner/key identity and signature bytes. This was a **historical consistency check against recorded test keys**, not a certification of physical ownership, production PKI, encryption or administrative isolation. The publication source check S4 did not repeat those six checks. Do not merge a hash of B with a semantic proof of E or C.

G. Contract finding, pending controls and next boundary

G.1. A correction with source obligations

A future owner-controlled version of the shared contract could name `DomainResult`, `PublishedEvidence` and `ConsumptionContext` explicitly. An independent examiner should construct its expected consumer context from actual producing `Work` output, exact received publication, provenance refs, authenticated saved `STATUS`, request/revision and time; it must compare the native consumer’s **actual** inputs and output to that independently built value. Validations should reject altered payload under a new signature, unrelated `Work` or review refs, stale `STATUS`, dropped fields and a candidate whose invented input agrees only with its own report. A small non-football numerical probe is an appropriate check of this generic seam before another expensive author. These are **future requirements**, not events inside G54B1. The subsequent G54C preparation, if documented elsewhere, must not be presented as a result of the first author or used to recolor G54B1’s frozen acceptance.

The old `INCOMPLETE` remains true even if later versions fix the defect. The next cold author needs separately pinned `kit/WHAT/examiner/source` versions and a distinct trial ID; a data variation should be frozen before seeing output. Maintainers have now learned from this case, so a later fresh model thread is not an independent untouched industry benchmark. Any eventual accepted trial should receive its own publication and status. Sources: `S1:§§7-8`; `S0:FINAL_RESULT.json /next_required_action`; owner publication directive §14.

G.2. Outstanding evidence, precisely named

Item	G54B1 recorded state	Why it matters
Frozen independent native acceptance	<code>FAIL_AT_ACTUAL_PREDICATE</code> , partial prefix at <code>venue_offer_consumption</code>	No full P01-P10/A-H score or acceptance.
Full adversarial matrix	<code>NOT_COMPLETED_AFTER_NATIVE_ACCEPTANCE_FAILURE</code>	Saved scalar, root-claim and supplied diagnostics do not stand in for it.
Runtime live Gemini role series	<code>NOT_RUN_NATIVE_ACCEPTANCE_PREREQUISITE_FAILED</code> ; <code>gemini_runtime_calls=0</code>	Authoring by an LLM and live semantic execution are distinct experiments.
Two final supplied proofs	<code>NOT_RUN_NATIVE_ACCEPTANCE_PREREQUISITE_FAILED</code>	The narrow actual-input diagnostics do not become final supplied acceptance.
Candidate-authored unit tests	<code>SOURCE_REVIEWED_NOT_EXECUTED</code>	Text of tests is neither test execution nor independent evidence.
Real-world booking/payment/cloud/QPU	Zero in reported trial	Only a local mock registry changed.
Old standalone RPC discrepancy	Preserved, unexplained, not retested here	A <code>scoped environments=[]</code> broker diagnostic does not resolve it.
Owner admission / Git publication during G54B1	No owner edits, commit or push	Publishing this historical capsule later is documentation admission, never source admission.

These values are from `S0:FINAL_RESULT.json`; no blank item is recast as zero. The visual case's useful execution and acceptance failure are both genuine recorded facts. A publication integrity check may verify an accurate history of an INCOMPLETE trial; it does not issue program admission.

G.3. Comparison and method without borrowed authority

The topic is the separation between creating a capability, exercising it under local rules and granting admission. [Voyager's authors](#) describe an accumulating executable skill library, curriculum and iterative feedback; it would be inaccurate to infer that every increase in skills necessarily expands operating-system permissions. [AutoGPT Forge's official CodeExecutorComponent documentation](#) describes shell execution controls and allowlist options; this likewise supplies no generic safety ranking. This appendix adopts claim-evidence-objection discipline from [CMU SEI's eliminative argumentation report](#) and artifact-description practice illustrated by the [USENIX OSDI '26 artifact call](#). None of those sources certifies Radiolaria, implies a conference acceptance, or substitutes for the case archive.

H. Evidence inspection and reproduction without candidate execution

H.1. Integrity and source map

Obtain the recorded S0 archive by its exact name and SHA-256 from a trusted copy. The archive contains 2,078 regular files, 2,077 top-level manifest rows and 40,628,730 uncompressed file bytes. S1 checked all manifest entries, paths, sizes, hashes and modes, no links/unsafe paths/duplicates, and gzip completion. S4 separately repeated archive identity and selected counts. This capsule includes a chosen source corpus, not a silent claim that it embeds all 2,078 records; `evidence/SOURCE_MAP.json` explicitly maps each included member. Its old `.py` files are stored as `.py.txt` with byte-identical contents and non-executable publication modes. Inspect files as data. Do **not** import candidate packages, replay embedded command strings, contact a venue, or make effects to validate the publication. `content_manifest.json` records the derived capsule postimages without self-hashing.

Start with `evidence/recorded/FINAL_RESULT.json`, `AUTHOR_TRIAL/freeze.json`, all four `AUTHOR_TRIAL/attempt_0{0..3}/FROZEN.json` and `REVIEW_DECISION.json`, followed by `AUTHOR_VISIBLE/HOW_TO_BUILD.md`, `TASK_VISIBLE/WHAT_TO_BUILD.md` and `EXAMINER_PRIVATE/examiner.py.txt`. Resolve the published local paths through `SOURCE_MAP`; check archive hashes against it. Read `evidence/trial_ledger.json` to distinguish predecessor trials, the G54B1 author cycle, and the fresh diagnostic. `evidence/CLAIMS_MAP.tsv` binds twelve public claims to exact original sources and method/limit. Neither map asks a reader to affirm the owner's interpretation without inspection.

H.2. Minimal read-only verification route

1. Verify SHA-256 and manifest coverage for S0; verify local `SOURCE_MAP` entries against included copied bytes. Do not treat an absent member as included merely because it existed in S0.
2. Compare the four candidate manifest hashes to four `FROZEN.json` records, link each `REVIEW_DECISION.json` to the next `feedback_input.json`, and distinguish completed confined runs from authored tests.
3. Read the final EXACT and SHIFT request/inventory and venue/requester reports. Enumerate permitted natural full-size intervals independently using half-open interval overlap. Check zero

complete exact candidates, the one explicitly approved revised combination, unchanged January 7/9 times and the integer sum.

- 4. Compare the final booking registry and readback bytes, three booking/repeat/verify receipt refs on each side, and the trusted parent observer’s total dispatch/effect returns. Read attempt 02’s registry and feedback separately; its incomplete evidence file is **not** a no-effect observation.
- 5. Decode E and C from the actual saved Work fields. Read B from the published body and its received copy. Compare B.offer to E, C.body to B, Work/review IDs to their real records, and independently reconstruct the status projection from B and saved STATUS. Avoid using C from the candidate as its own oracle.
- 6. Read HOW_TO_BUILD.md:160-169,209-210, examiner.py:155-169 and the final correction note. Keep **author-reported contradiction** and **independent reviewer finding** separate. S1 contains a previous selective cryptographic and scalar verification; this publication check does not claim a new native, live or signature run unless accompanied by its own recorded outputs.

H.3. Independently deriving the consumer’s status projection

A reader can check the decisive relation without executing candidate code. Open final publisher/evidence/{exact,shift}/offer_body.json as B, requester/evidence/{exact,shift}/status.json as a saved signed status record, and requester/evidence/{exact,shift}/requester_work.json as actual Work. Decode only the latter file’s inputs.offer_json to obtain C. For each of EXACT and SHIFT, construct the expected projection from B and STATUS fields as follows; do not read the desired values from C while constructing it.

C.source_projection field	Independently selected source
observed_at, pt, ttl, valid_from, valid_to	B’s time_envelope.source_observed_at, pt_created_at, ttl_seconds, valid_from, valid_to respectively.
offer_revision, schedule_revision	B’s offer.offer_revision, offer.schedule_revision.
publisher, recipient	B’s own publisher, recipient.
request_revision	STATUS value.request_revision; also cross-check against B’s bound request revision.
status, status_checked, status_until	STATUS value.entry.state, value.checked_at, value.valid_until.

Both reconstructed dictionaries equal actual C.source_projection; C.body equals B; B’s offer equals the **decoded** producer outputs.offer_json. B’s source_work_ref and source_review_ref are checked against their saved real Work artifact and review decision, so the publication cannot merely invent plausible values for those fields. Compare the exact body bytes at publication and fetch as a separate test. The recorded status signature check is in S1; dictionary equality alone does not authenticate STATUS. This read-only derivation reproduces the relevant *relationship*, not native admission, full protocol replay or cryptographic identity of a real operator.

H.4. Questions for an independent reader

What exactly was computed and changed, and which saved bytes and observer support it? Which actor could change candidate, kernel, examiner and admission? At what causal moment was each field of E, B and C available? What competing explanation remains for the observed failure? Which Gate 5 obligations are unrun in **G54B1**, and which were merely diagnostics? A reader may dispute an architectural interpretation while preserving the recorded execution, failure and exact source bindings.

H.5. Selected primary-source identity index

The full inventory, including each attempt and recorded trace, is [SOURCE_MAP.json](#). This smaller index lets a reader check the highest-impact original bytes quickly. Q/ abbreviates SUPERVISOR_EVIDENCE/p7_candidate_1244df5d/; the abbreviation is only for this table. The S1 review is an independently authored external document, not an S0 member.

Original source (relative to S0, except S1)	SHA-256
FINAL_RESULT.json	af41bbef5307e8aab6cf24d097f0de5d3c6e8c2322e8d86ceefd4adce1737b5d
AUTHOR_VISIBLE/HOW_TO_BUILD.md	6636167b75d8fec5786e80cb5ce3027c02748d58d5d63765655b1e451d60dea3
TASK_VISIBLE/WHAT_TO_BUILD.md	dbcdfeeaca9405687dd30bba33a72a510729d606c7d99ee52a5f390d426963a1
EXAMINER_PRIVATE/examiner.py	874b4b7a82c15bc2212404ccef182519ef16681bb9a85b0addf6f6f336ce2372
Q/worker/publisher/evidence/shift/venue_work.json	887e3b4feecb3ef72450d5b219797309cfdb74b3ddbc8b1697b3bbbd31c99d7d
Q/worker/publisher/evidence/shift/offer_body.json	b92ee183c725b2052e29cd1840d195fb3c5bc6b0d78984a1d1339f5285e61ab3
Q/worker/requester/evidence/shift/requester_work.json	8cd81376a262512a1a8dd3d925829d6e94bcab13621570cbda7eb7ab3edb0a64
Q/worker/requester/evidence/shift/status.json	ea463b7c3c507824a4ca33553257010a427e13ad49c8dbc9fbb609a9eb6a3d66
Q/trusted_parent_receipt.json	35cfc81c0af11b958c33a56710bb17d90eb641f408703aca023cf883c22de99a
Q/worker/publisher/state/registry.json	1e0d2d310d7032ae3a8471e771f54cf6dabc1616682237527a763c94361d7300
S1 independent review	87152f7f44df0aa3995fd4edd960e44f55c086368abbb133d23607fa6b543e18

Final provenance statement. This appendix describes a historical candidate that performed useful controlled work and revealed a specific frozen contract mismatch. It is a documentation artifact about an **INCOMPLETE** trial. The rejected football candidate is not installed, admitted, or silently upgraded by this publication.